

Japanese Magic Squares

The history of Japanese mathematics is particularly fascinating. In *A History of Japanese Mathematics*, originally published in 1914, the historians David Eugene Smith and Yoshio Mikami wrote that Japanese math had historically possessed a “genius for taking infinite pains” and “ingenuity in untangling minute knots and thousands of them.” On the one hand, mathematics uncovers absolute truths that should not vary between times and cultures. On the other hand, the types of problems that distinct groups tend to focus on and their idiosyncratic approaches to them, not to mention differences in notation and communication, provide great scope for noteworthy cultural differences, even in a field as austere as math.

Creating the Luo Shu Square in Python

Japanese mathematicians had a fondness for geometry, and many of their ancient manuscripts pose and solve problems related to finding the areas of exotic shapes like circles inscribed within ellipses and Japanese hand fans. Another steady area of focus for Japanese mathematicians throughout several centuries was the study of magic squares.

A *magic square* is an array of unique, consecutive natural numbers such that all rows, all columns, and both of the main diagonals have the same sum. Magic squares can be any size. [Table 2-9](#) shows an example of a 3×3 magic square.

Table 2-9: *The Luo Shu Square*

4	9	2
3	5	7
8	1	6

In this square, each row, each column, and both main diagonals sum to 15. This is more than just a random example—it’s the famous *Luo Shu square*. According to an ancient Chinese legend, this magic square was first seen inscribed on the back of a magical turtle who came out of a river in response to the prayers and sacrifices of a suffering people. In addition to the definitional pattern that each row, column, and diagonal sums to 15, there are a few other patterns. For example, the outer ring of numbers